

Decision Quality Is Not Model Quality

A certified environment that measures decisions, not predictions — exact replay, proven attribution, and exact counterfactual reward over a capacitated time-graph, built to study the lever that actually moves outcomes under hard constraints.

Status. This is the *front door* to the work: it leads with the built artifact and the four properties it proves, states the one measured characterization with its noise, and is honest about where the program is still a frame. The formal development — definitions, the recurrence, the bounds, the proofs — lives in the frozen ten-section paper (paper-assets/paper/), which this note links to and does **not** duplicate.

Honesty labels are load-bearing and used exactly as the source apparatus tags them: **[PROVEN]** = backed by a test, golden, or benchmark; **[CHARACTERIZATION]** = a measured fact about one specific regime, with its noise stated — *not* a law; **[FRAME / OPEN]** = a research direction, argued but not built or proven. A frame is never promoted to a result because it reads better.

Vocabulary is domain-neutral by construction: *flow units* traverse *nodes* over *lanes* by boarding *connections* (scheduled, capacity-limited departures). The same result reads for any capacitated time-graph.

1. Thesis — decision quality is not model quality

A system that fails under hard constraints rarely fails because its predictions were imprecise. It fails because the wrong things were *committed*: admitted when they should have been refused, solved when the answer no longer bound, re-planned too late to matter. The quantity that moves outcomes is the **quality of the decision**, not the fidelity of the model the decision consulted — value of information is worth only what it changes in the choice [Howard 1966].

The constructive claim of this note is that the distinction can be made **measurable**. Below is an apparatus that, over a capacitated time-graph, says what happened, whether it was avoidable, whose fault it was, and what a decision was worth — each backed by a proof or an exact replay rather than an estimate (§2–§4). On that apparatus the prediction / decision distinction stops being rhetoric and becomes a number with a certificate (§5).

The larger claim — that **the dominant lever is the decision of what to hand the optimizer, not the optimizer itself** — is named here as a frame and developed in §6 **[FRAME / OPEN]**. The analogy that first motivated the conjecture (a neighbouring field in which the pipeline stopped being a perception problem and became a decision problem end-to-end) is told in the notebook, not argued here → Notebook: “The day driving stopped being a perception problem.”

2. The apparatus

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The artifact is not a planner; it is the machinery *around* a planner that certifies what the planner did. One engine, three responsibility layers:

- **The world (DES).** A capacitated time-graph replayed by a discrete-event engine in strict time order, with capacity-blocked flow units entering per-lane overflow queues under a **total order** (`priority_class`, `enqueued_atu`, `flow_unit_id`). All time is integer; capacities are integer; the run is reproducible to the byte. This is the *mechanism* — the physics that must be exact and fast — and it is compiled.
- **The certifier.** The proofs *about* what the mechanism did: a capacity-free, destination-rooted feasibility oracle (earliest on-time arrival, ignoring capacity); a typed reason taxonomy (**R1 world / R2 envelope / R3 algorithm**) that maps every non-ideal outcome with **zero unexplained mass**; and a certificate lattice of upper bounds, $\text{oracle} \supseteq \text{cohort-LP} \supseteq \text{achieved by construction}$.
- **The seams.** Six declared, stable interfaces (World, Gate, Propose, Imagine, Navigate, Optimize), populated today by non-learned baselines and graded by the certifier. A policy swapped into one seam cannot silently change another’s semantics.

The price of time is now part of the certified apparatus. The world measures *held time* — held-but-not-in-transit — as a deterministic integer field (`held_atu`, the strict superset of queue-wait: it counts scheduled

dwell even where a unit was never refused, so a unit that waits for a departure it always makes is no longer counted as free). The certifier prices that field into realized value through a convex, domain-parametrized wait cost $C_{\text{wait}}(w) = v \cdot (a \cdot w + b \cdot w^2)$ (a `WaitCostProfile`), whose convex term $b > 0$ is load-bearing: it is what couples one commitment’s value to another’s. All three planes — the world that *realizes* `held_atu`, the substrate that *certifies* the priced value, and the agent that *optimizes* it — agree on the wait-adjusted value **to the bit** (a three-plane round-trip). The four guarantees below are unchanged by this; the pricing is an addition to the certified surface, not a revision of it.

The split is deliberate: the mechanism must be *trusted and fast*, the judgment *expressive and frequently revised*, so the deterministic kernel is compiled and the proofs sit in a higher-level layer above it. The design is an instance of the **certifying-algorithm** discipline — an algorithm that returns, with its answer, a witness an independent checker can verify [McConnell et al. 2011] — the posture that makes machine-checked guarantees about combinatorial outputs tractable [Abdulaziz et al. 2019]. (*Full directory-level map: §2 of the frozen paper, sections/02-apparatus.md.*)

3. What it guarantees

PROVEN

Four properties, each test- or golden-backed:

1. **Exact replay.** Same seed $\Rightarrow$ byte-identical run — KPIs, records, surfaces, manifest. Verified by the golden worlds and, at scale, by `replay_identical = True` at every point on the benchmark curve. Determinism is a *correctness property* of the kernel, not a statistical aspiration.
2. **Typed, proven attribution.** Every non-ideal outcome is classified world’s-fault (genuinely infeasible — no path exists at any release) versus policy’s-fault (feasible-on-time, but refused; or feasible-late) by the destination-rooted oracle and the R1/R2/R3 taxonomy, with zero unexplained mass. “The network was full” becomes a *theorem about the schedule*, not an excuse. The temporal-feasibility substrate is a single time-sorted scan over connections [Dibbelt et al. 2018]; the lattice’s flow-unit-count-invariant upper bound descends from multicommodity-flow relaxation [Garg & Könemann 2007].
3. **Exact counterfactual reward.** Because the world replays deterministically, a paired same-seed rollout (act vs. no-op) yields the *true* marginal value of an action — the reward is certified by construction, not estimated — and cumulative reward telescopes exactly to $V(\text{selected_set})$. The reward is never on a gradient path; it is a replayed float.
4. **Regret against a certified oracle.** Any policy is graded by **budgeted regret** $V(S^*) - V(S_{\text{policy}})$ against a (cost, budget)-constrained certified ceiling, so “did the policy improve” is a proven quantity, not an inference.
5. **Coherent priced value.** The wait-adjusted value the agent optimizes equals what the substrate certifies equals what the engine realizes — bit-exact, the three-plane round-trip on `held_atu`-priced value. This is what makes any *wait-aware* regret measurement trustworthy: the number a policy is scored on is the number the world actually produced, down to the last integer of held time.

(*Definitions, the lattice sandwich, and the refusal-certificate theorem: §3–§5 of the frozen paper.*)

4. Performance — the proof pays for itself

PROVEN

Producing a *certified* plan is not a tax over routing; on the measured configuration it is **cheaper** than the capacity-blind baseline. All numbers below are **absolute, single-machine** measurements on **one scenario family** (backbone topology / hourly cadence / loose capacity, seed 42, Apple M2 Pro); every “ $\times$ ” is strictly *this apparatus’s certified planner versus this apparatus’s own Dijkstra baseline*, **not** a claim against any external system, and **not** a topology/cadence/capacity sweep.

- **Stepping the world is cheap.** A 1,000,000-flow-unit world — 8.86M discrete events — replays in 3.07 s ($\approx$ 2.88M events/s); throughput *rises* with scale as fixed costs amortize. A full paired counterfactual at that scale is 6 s of wall.
- **Certification is cheaper than the capacity-blind control.** At 1M the certified time-dependent A^* runs at 26.8 μ s per flow unit versus 62.0 μ s for the Dijkstra baseline — **2.31 $\times$ faster on this configuration, with the gap widening as scale grows**. The reason is structural: the capacity-free destination oracle that makes the plan *certified* is the very same **admissible heuristic** that prunes the A^* search to a handful of labels per flow unit. You do not trade speed for the proof — *the proof pays for itself*.

- **The proof bound is constant.** Decomposed at 1M, the certify cost is $\approx A^* 58\% \cdot \text{apply } 18\% \cdot \text{ledger } 14\% \cdot \text{DES } 6\% \cdot \text{attribution } 3\% \cdot \text{oracle } <1\%$. The oracle is built per *destination*, not per flow unit, so it is 0.3 s at every scale — the proof part does not grow with the problem; the guided search it enables does. Turning a realized run into a typed, proven three-way split costs 3%.
- **Exactness is (almost) free.** Byte-identical replay holds at every scale; the only price is materializing and hashing records ($\leq 44\%$ overhead at 1M, one-time, off the hot loop), at 2.2 GB peak memory.

Pricing wait costs nothing here: it adds one deterministic integer field (`held_atu`) to the record and a pure function over it, off the hot loop — the throughput figures above are unchanged by it.

(Full tables and method: `paper-assets/paper/sections/10-reproducibility.md` and the benchmark source.)

5. Estimation is not decision

(CHARACTERIZATION — REGIME-SPECIFIC)

On this apparatus the prediction / decision distinction is not asserted but *measured* — and the measurement is reported with its scope, because it is a fact about an **easy regime**, not a law. In the currently generated worlds, a **3.5× spread in estimator error moves the committed decision by less than noise**: top-2 budgeted-ranking regret is identically 0 across every estimator (including a non-value severity baseline), and at top-1 the estimators do not separate beyond noise — mean regret 82.6 versus 103.5 over 72 suites, a gap far smaller than the across-suite **standard deviation** (203.5 and 210.5), with overlapping 95% confidence intervals (± 47.0 and ± 48.6). This is exactly what theory predicts where the decision-optimal predictor and the accuracy-optimal predictor come apart [Elmachtoub & Grigas 2022; Donti et al. 2017; Howard 1966], and it echoes empirically the decision-focused finding that accuracy is not a reliable proxy for decision utility [Wilder et al. 2019].

The cause is **over-determination**: realized per-epoch value is dominated by 2 interventions, so any top-k ranking already reaches the optimum and estimator MAE cannot move the committed set. This must not be conflated with a *recourse* claim: the over-determined value distribution and the §8.4 partial-recourse regime are **two separate axes** — the decoupling here holds independent of any recourse assumption, and an earlier draft that attributed it to §8.4 was a conflation, corrected on the record. The honest boundary on the finding is the regime where prescription quality *does* couple to predictive-weight quality [Bertsimas & Kallus 2020]: that boundary condition is where this characterization stops, and it is named, not hidden.

This characterization is now one end of a **measured axis**, not an isolated finding. It is the *flat* end: where value is over-determined and waiting is cheap (the `slack_elastic` regime — read as mid-mile-transport-like), the committed decision barely moves, so model quality is nearly irrelevant. The *coupled* end — where waiting is convex and capacity is scarce (the `steep_scarce` regime), and the decision lever is real and large — is the new §6 result. Both ends are measured; the thesis is shown **vacuous where coupling is absent and real where it is present**, which is a stronger and more honest claim than either end alone.

(Framed as a consequence of the certified executor working — §2–§4 above — never as a standalone headline.)

6. The decision-above-optimization frame

(FRAME · OPEN)

This section is where the frame meets its first measurements. Two kinds of claim now live here at once, marked inline so neither is mistaken for the other: **proven existence results at a certified scope** ([`proven`]) and the **general dominance law they instantiate but do not establish** ([`frame`]). A frame is never promoted to a result because a measurement made it tempting — the labels hold in the prose *and* in the mark. The thesis the apparatus exists to test: in constrained dynamic systems the dominant lever is **not the optimizer but the decision of what to hand it** — which flow units to commit, which to refuse, what envelope to assume, and at which points in time to (re-)solve at all. Optimization becomes a *banded tool invoked at chosen decision points*, not the top-level loop; the **committed set and the assumptions** are the decision variables, and budgeted regret against the certified oracle is the score. This is the OR/RL bridge made explicit — a policy over a capacitated-time-graph decision process [Sutton & Barto 2018], with rollout and receding-horizon resolution as the principled lookahead [Bertsekas 2019], and machine learning placed *at the discrete decision* inside the solver rather than replacing it [Bengio et al. 2021]. The neighbouring analogy — pipelines reorganized so every module serves the terminal decision, and in the limit collapsed into one end-to-end map — runs *with* the frame in its planning-oriented form [Hu et al. 2023, UniAD] and *against* it in its monolithic form [Hwang et al.

2025, EMMA]: the contrast sharpens the claim that the lever is the discrete decision layer, not a larger learned optimizer.

What was open, and is now measured. The apparatus was built to make this frame *provable*; the sprint that priced held time (§2) turned three of its sentences from conjecture into measurement. Stated with the marks:

- **The competing-bottleneck worlds now exist.** A *controlled bottleneck* — windowed demand over capacity on a named cut — manufactures tunable, sub-additive contention that preserves the engine’s invariants and self-reports its regime. What §5’s generators could not produce, this now produces on demand. [proven]
- **The decision gap is measured.** Regret against the conservative certified oracle grows monotonically with the cost of time: R_{raw} rises $2.53 \rightarrow 4.52 \rightarrow 9.88 \rightarrow 24.82$ as steepness s goes $0 \rightarrow 0.75$, monotone *per seed*, the confidence interval excluding zero from $s = 0.25$, with the falsification control flat at zero. [proven] (ILP-certifiable scale $n=12$, $d=120$, HiGHS-certified; controlled bottleneck $\rho=2.0$ — the natural world is barely congested at this scale; greedy = `CapacityAwarePlanner(value_weighted)`; conservative lower-bound oracle.) Inside that gap is the **inflicted-wait externality** — value recoverable only by anticipating the wait a commitment inflicts on units committed *later*, invisible to any single-unit marginal — and it is a guaranteed-real lower bound: $R_{\text{wait}} = +9.63$ $[+5.4, +13.8]$ (domain-agnostic, committed basis) and $+13.77$ $[+8.1, +19.5]$ (steep_scarce, admission-enforced). [proven] (same scope; the oracle is conservative by construction — a subset feasible region and an inner piecewise-linear wait cost — so every positive regret is a floor on the true gap, never an inflation of it.) What stays open is the **general** claim: that deciding-before-solving dominates *any* optimizer on *any* world. A lower-bound oracle can certify existence at a scope; it structurally cannot certify a universal law — so that remains the frame the apparatus was built to test, now instantiated with a measured magnitude but not proven. [frame]
- **The learning target now exists; the learner does not.** Define *greedy-wait* as the Level-1 policy that builds the committed set one unit at a time, each step committing the highest own-wait-adjusted marginal $\Delta_j = V_{\text{realized}}(S \cup \{j\}) - V_{\text{realized}}(S)$ by certified rollout — myopic, blind to the wait each commitment inflicts on units not yet in S . Under *free* waiting that blindness cost nothing and greedy-wait was optimal — which is precisely *why* the earlier note could honestly report no learning target. Under *convex coupled* waiting it is no longer optimal: the inflicted-wait residual $R_{\text{wait}} > 0$ is a certified, measured target. [proven] The method that would close it is not built. [frame] And closure is framed **tractable-first**, not as a foregone learned policy: (a) characterize why greedy-wait leaves the gap; (b) a tractable, non-learned anticipatory *set-search* may close it outright — which would **strengthen** the thesis, since the lever would then be a cheap decision method, not a heavy optimizer; (c) learning enters only after that, and only as a value *surrogate* that amortizes the exact certified rollout when the non-learned search closes the gap but is too slow at scale — the regime where carried predictions would need the consistency/robustness guarantees of the algorithms-with-predictions kind [Mitzenmacher & Vassilvitskii 2022; Purohit et al. 2018; Lykouris & Vassilvitskii 2021]. This is Level-1 *set-selection* — which units to commit, refuse, or defer — **not** routing inside a fixed set (conflict-based search / multi-agent RL), which is the distinct, NP-hard Level-2 problem; treating one as the other is a category error.

The **Imagine seam** is the open direction with the cleanest honest contrast, and the sprint sharpened it rather than closed it. The artifact is *already* a world model — of the **certified-counterfactual** kind: an exact, replayable environment whose rewards are proven, not predicted. R_{wait} is the first *certified* quantity a learned imagination would have to target. But two things must stay apart: closing *present-epoch* inflicted wait is the Level-1 set-selection ladder above; a *temporal, multi-epoch* forecast that reads contention several hops ahead to decide *when* to admit is a richer, later regime. Both are open. [frame] The references that frame this seam each carry the contrast in-line: a learned generative world model trained to dream rollouts [Ha & Schmidhuber 2018]; planning with a *learned, approximate* model [Schrittwieser et al. 2020, MuZero]; and a *proposed, unbuilt* architecture for prediction in representation space [LeCun 2022, JEPa] resting on an energy formalism [LeCun et al. 2006]. Each is learned, approximate, uncertified, or merely proposed — against this environment’s exact, certified, deterministically-replayable ground truth. None of it is built here; it is the seam still to come, told in the notebook → Notebook: “The world model we have, and the one we don’t.”

7. Related work — positioned, not listed

The work sits at the confluence of three lines, and earns its place by *positioning* against them rather than enumerating them. From **certifying algorithms** it takes the witness-plus-checker posture [McConnell et al. 2011; Abdulaziz et al. 2019] and applies it to *failure attribution*, not just to a single output. From **scheduled-network feasibility** it takes the time-sorted scan [Dibbelt et al. 2018] and the multicommodity-flow relaxation

[Garg & Könemann 2007] and uses them as a *certifying layer* beneath attribution, not as the contribution. From **decision-focused learning** [Elmachtoub & Grigas 2022; Wilder et al. 2019; Donti et al. 2017] it takes the prediction/decision separation and supplies the *inverse* boundary — the over-determined regime where the separation collapses (§5), with [Bertsimas & Kallus 2020] marking where it re-couples.

Two further anchors mark the boundary the new §6 result crosses. From **submodular optimization** it takes the *ancestry* of the whole claim: that greedy’s optimality is **conditional on the structure of value** — under monotone submodularity marginal-gain selection reaches a $(1 - 1/e)$ fraction of the optimum, near-optimal but **not exact** [Nemhauser et al. 1978]. The **separability boundary** this note draws is a sharper, adjacent condition, and it is our own result, not theirs: greedy-wait equals the optimum *iff* each unit’s held time is independent of the committed set ($w_j \perp S$), and §6 exhibits a regime where that independence fails and *measures* the residual it leaves. That residual is near-optimality slack plus separability myopia — **not** a submodularity violation; the inflicted-wait externality itself carries the diminishing-returns sign. From **ex-ante admission control** it takes the operational form of the lever — that admitting the right load *before* commitment can outperform reactive control under overload [Zhang et al. 2022] — and supplies what a systems benchmark cannot: a certified regret measurement of that decision against an optimal-recourse oracle.

One residual is left explicitly open. A *primary* method for counterfactual-credit-assignment-for-agents — one that separates agent-caused loss from world-imposed loss as a methodology, not a survey — has no anchor here beyond a single survey of the young field [Yehudai et al. 2025]. This note **does not fill that slot** and does not manufacture a citation for it; the natural fill (regret-decomposition against a certified optimal-recourse oracle, using this environment’s exact counterfactual as the baseline) is described as a direction, not asserted as prior art.

8. Reproducibility

PROVEN

The apparatus is **Apache-2.0** and reproduces from a clean clone against **committed golden data** at a **pinned tessar-core tag**: deterministic counts and digests are **byte-asserted** against committed expected tables, while wall-times, throughput, and memory are *reported* with the hardware stated (Apple M2 Pro, 10 cores; single-threaded planner over the compiled kernel) — never byte-asserted. A per-run manifest embeds the wire-protocol fingerprint, so a run under a different protocol is mechanically distinguishable. Where a result’s raw run folder is not retained, the reproduction regenerates the *committed extract* and says so in its output, rather than claiming a from-zero re-derivation. The refusal-certificate invariant is re-checked every run — a positive count is a certification-bug detector, not a warning to ignore.

The wait-pricing sprint adds two reproduce targets under the same discipline: the **decision-gap sweep** (R_{raw} versus steepness, monotone per seed with its confidence interval) and the **inflicted-wait decomposition** (R_{wait} as a conservative lower bound). Both reproduce the committed golden extract byte-for-byte; the raw sweep runs are not retained, and the output says so.

(One-command reproduction and the invariant-as-bug-detector methods box: §10 of the frozen paper.)

9. Limitations & open

Stated plainly, because the honesty of the negative space is the point:

- **Scope of every number.** Single-machine, one scenario family, seed 42; “faster” is versus the apparatus’s own capacity-blind baseline. Whether the same ratios hold across topologies, saturation regimes, or hardware is **not** claimed — it would need the sweep this benchmark does not run.
- **The decision finding is regime-bounded.** The estimation/decision decoupling (§5) is a **characterization** of an over-determined regime, not a general law; its boundary is named, not hidden.
- **A certified oracle and natural congestion cannot yet coexist at scale.** The congested-regime result is real but scope-bound by a hard tension: a *certified* oracle and *large-scale natural* congestion are mutually exclusive at current solver capability — the ILP certifies at $n = 12$, not at production scale. So the gap is demonstrated at the certifiable scale with an *injected* controlled bottleneck; production-scale natural congestion remains open. [frame] (This replaces the earlier, now-false claim that the congested regime was unbuilt: it is built and measured — at a named, small scale.)
- **The target is proven; the learner is not.** Under convex coupled wait, greedy-marginal is no longer optimal and the inflicted-wait residual is a certified learning target (§6). [proven] No learned agent is built against

it, and the method that would close it is deliberately framed tractable-first, not as a foregone learned policy.

`[[frame]]` The honest statement flipped from “no target” to “target proven, learner unbuilt.”

- **The priced-value lattice has a located open rung.** The certificate lattice extends to priced value at the oracle rung always, and at the LP rung under hard tolerance; the convex-LP middle rung under *linear* tolerance (a convex relaxation sandwiching priced value) is a located open problem, not a silent gap. `[[frame]]`

Hand-off. This note is the artifact-first front door. The formal depth — the capacitated-time-graph definitions, the four-rung certificate lattice and its sandwich bound, the refusal-certificate theorem, the commit-set lever, the loop neutralization, the scope conditions — is the frozen ten-section paper (paper-assets/paper/sections/01..10 + main.md). Read this for *what was built and why it matters*; read the paper for *why it is true*.

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